

Low-light image enhancement based on Retinex theory and dual-tree complex wavelet transform^{*}

YANG Mao-xiang (杨茂祥)¹, **TANG Gui-jin** (唐贵进)^{1**}, **LIU Xiao-hua** (刘小花)¹, **WANG Li-qian** (王力谦)¹, **CUI Zi-guan** (崔子冠)¹, and **LUO Su-huai** (罗苏淮)²

1. Jiangsu Key Lab on Image Processing & Image Communication, Nanjing University of Posts and Telecommunications, Nanjing 210003, China

2. School of Electrical Engineering and Computing, University of Newcastle, NSW 2308, Australia

(Received 28 March 2018; Revised 28 April 2018)

©Tianjin University of Technology and Springer-Verlag GmbH Germany, part of Springer Nature 2018

In order to enhance the contrast of low-light images and reduce noise in them, we propose an image enhancement method based on Retinex theory and dual-tree complex wavelet transform (DT-CWT). The method first converts an image from the RGB color space to the HSV color space and decomposes the V-channel by dual-tree complex wavelet transform. Next, an improved local adaptive tone mapping method is applied to process the low frequency components of the image, and a soft threshold denoising algorithm is used to denoise the high frequency components of the image. Then, the V-channel is rebuilt and the contrast is adjusted using white balance method. Finally, the processed image is converted back into the RGB color space as the enhanced result. Experimental results show that the proposed method can effectively improve the performance in terms of contrast enhancement, noise reduction and color reproduction.

Document code: A **Article ID:** 1673-1905(2018)06-0470-6

DOI <https://doi.org/10.1007/s11801-018-8046-5>

Image enhancement is an important branch of digital image processing. The purpose of image enhancement technology is to improve the quality of the image to achieve satisfactory results. Images taken under low-light conditions often result in a narrower dynamic range and lower contrast. So far, researchers have proposed a lot of good image enhancement algorithms. In 2015, Pooja^[1] developed a method to enhance the quality of image and contrast. In 2016, Cheolkon Jung et al^[2] proposed the enhancement of readability of low-light images based on dual-tree complex wavelet transform (DT-CWT) while achieving noise reduction and color reproduction. In 2017, Seonhee Park et al^[3] proposed an optimization-based low-light image enhancement method using spatially adaptive I_2 -norm based Retinex model. The proposed method can preserve details of bright areas. Karumuri^[4] combined the edge based weighted scheme along with guided image filtering, and proposed an interesting algorithm-weighted guided image filtering scheme. Kyungil et al^[5] proposed a new single image enhancement method based on a mixture of dark channel prior and contrast limited adaptive histogram equalization with discrete wavelet transform algorithms. Guo et al^[6] proposed a simple and effective low-light image

enhancement method, which intended to enhance a low-light image by estimating its illumination map. Although the above algorithms have improved the visual effect of low light images to some extent, it is difficult to achieve color reproduction in low-light images.

In this paper, we propose a method based on Retinex theory to enhance illuminance in a scene with low-light or non-uniform illumination, while enhancing the contrast based on dual-tree complex wavelet transform and achieving noise reduction and color reproduction. Experimental results show that the proposed method outperforms some existing methods in the subjective and objective evaluation.

Retinex theory is a common method of image enhancement based on scientific experiments and scientific analysis. Jobson et al proposed single-scale Retinex (SSR)^[7] algorithm and multi-scale Retinex (MSR)^[8] algorithm based on the center-surrounded Retinex algorithm. SSR algorithm is described as follows:

$$R_i(x, y) = \log I_i(x, y) - \log [F(x, y) * I_i(x, y)], \quad (1)$$

where x and y are the pixel coordinates in an image, $I(x, y)$ is an input image, R_i denotes the reflectivity image of the i -th color channel, $*$ denotes the convolution, and $F(x, y)$ is the Gaussian surrounding function with the specific formula:

^{*} This work has been supported in part by the National Natural Science Foundation of China (Nos.61602257 and 61501260), the Natural Science Foundation of Jiangsu Province (No.BK20160904), the Postgraduate Research & Practice Innovation Program of Jiangsu Province (No.KYCX17_0776), the Natural Science Foundation of the Higher Education Institutions of Jiangsu Province (No.16KJB520035), and the NUPTSF (Nos.NY214039 and NY215033).

^{**} E-mail: tanggj@njupt.edu.cn

$$F(x, y) = Ke^{-(x^2+y^2)/c^2}, \quad (2)$$

where K is a constant matrix, and c is a scale constant. A small scale constant produces a good dynamic range compression, but results in poor color reproducibility. Conversely, a large scale constant produces good color reproduction, but results in not well compressed dynamic range. The MSR algorithm is described as follows:

$$R_{MSR}(x, y) = \sum_{n=1}^N w_n R_{n1}(x, y), \quad (3)$$

where N is the total number of scale parameters, and w_n is the weight parameter of the image processed by the n -th scale parameter.

The purpose of the MSR algorithm is to reduce the halo artifacts on high-contrast edges and maintain the balance with dynamic range compression and color reproduction. MSR produces good dynamic range

compression, but still suffers from halo artifact.

In Eq.(1), the input image $I(x, y)$ is not smooth at the edge, where the luminance is quite different, causing the halo artifacts which make the result unnatural. These artifacts can be reduced by using edge-preserving filters. The current common method is to use the edge-preserving bilateral filter instead of the Gaussian filter as a surround function to estimate the illumination component. Although the bilateral filter preserves the detail information well, the time complexity is high. Therefore, we use the guided filtering^[9] instead of the Gaussian filter in the Retinex algorithm, which also has the function of smoothing and preserving the edge details. The workflow of the proposed algorithm is depicted in Fig.1.

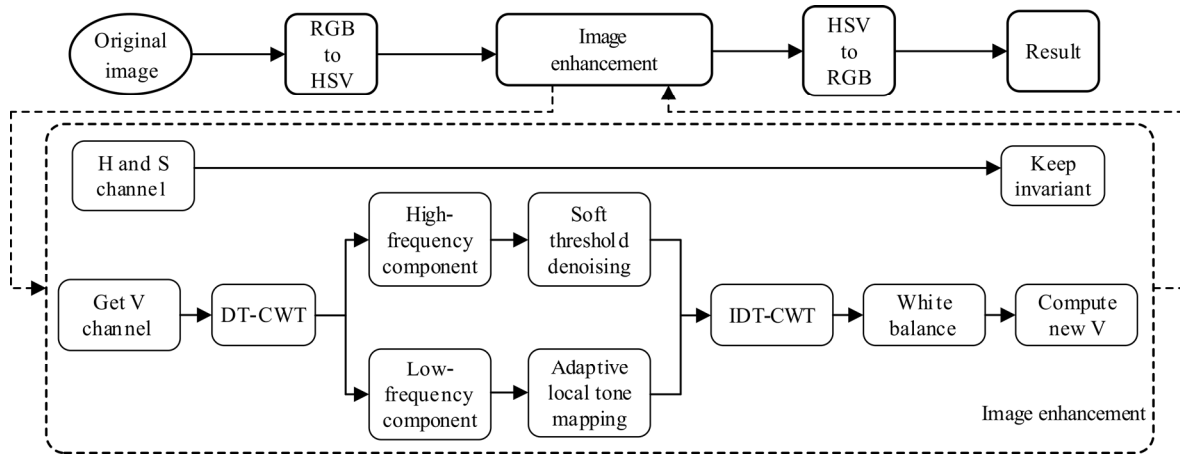


Fig.1 The workflow of the proposed algorithm

We convert low-light images from the RGB color space to the HSV color space. Since the HSV color space is closer to the perception of color in the human visual system, it is convenient for the processing and recognition of the color information. We process an image in the V-channel of the HSV color space.

Since DT-CWT^[10] can significantly improve the translational sensitivity and DWT has good performance in terms of noise reduction, we use DT-CWT for low-light images to improve contrast enhancement.

Dual-tree complex wavelet transform operates through two parallel real filters group. One-dimensional dual-tree complex wavelet transform is shown as Fig.2, where tree a and tree b use different filter banks to represent the real and imaginary parts of the complex wavelet, respectively. Since the image is a two-dimensional signal, when the image is subjected to a two-dimensional dual-tree complex wavelet transform, the method is similar to the two-dimensional discrete wavelet transform. Firstly, one-dimensional dual-tree complex wavelet transform is performed on the image rows, and then is performed on the image columns. It decomposes the image into two low-pass sub-images and six high-pass sub-bands. It

reduces the fuzzy artifacts in the decomposition process.

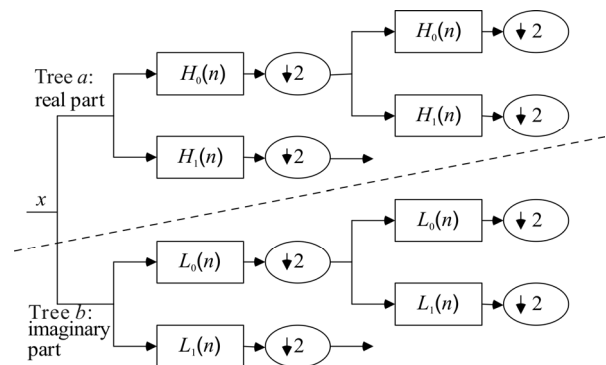


Fig.2 1D dual-tree complex wavelet transform

Since wavelet theory has low entropy and multi-resolution features, noise reduction can be achieved by processing wavelet coefficients. Therefore, we use DT-CWT to process low-light images in the wavelet domain. DT-CWT decomposes an image into low-pass and high-pass sub-bands. Since the noise mainly appears in the high-pass coefficients, we perform different contrast enhancement and denoising in the low-pass sub-bands and

the high-pass sub-bands.

We use the improved local adaptive tone mapping for contrast enhancement in the low-pass sub-band. According to the Weber-Fechner law, the human visual system approximates brightness perception as a logarithmic function. In order to compress the global dynamic range of the image, we use the function proposed in Ref.[11]:

$$L_g(x, y) = \frac{\log(L_w(x, y) / \bar{L}_w + 1)}{\log(L_{wmax} / \bar{L}_w + 1)}, \quad (4)$$

where $L_g(x, y)$ is the global luminance output, $L_w(x, y)$ is the luminance value of the input image, L_{wmax} is the maximum luminance value of input, and $\bar{L}_w$ is the logarithmic mean luminance which is as follows:

$$\bar{L}_w = \exp\left(\frac{1}{N} \sum_{xy} \log(\delta + L_w(x, y))\right), \quad (5)$$

where N is the total number of pixels in an image and δ is a constant to avoid the occurrence of singularities if there are black pixels in the image.

The local adaptive algorithm based on Retinex theory is applied after the global adaptation process. By introducing the edge preserving filter, we can reduce the artifacts. We use the guided filter instead of the Gaussian filter in the Retinex algorithm. The local adaptation equation is as follows:

$$L_l(x, y) = \log L_g(x, y) - \log H_g(x, y), \quad (6)$$

where $L_l(x, y)$ denotes the locally adapted output and $H_g(x, y)$ is the output of $L_g(x, y)$ applied to the guided filter.

$$H_g(x, y) = \frac{1}{|\omega|} \sum_{(\varepsilon_x, \varepsilon_y) \in \omega(x, y)} [a(\varepsilon_x, \varepsilon_y) L_g(x, y) + b(\varepsilon_x, \varepsilon_y)], \quad (7)$$

where $\varepsilon_x, \varepsilon_y$ are the coordinates of neighborhood pixels, and $\omega(x, y)$ is a square window with radius r and center (x, y) , $a(\varepsilon_x, \varepsilon_y)$ and $b(\varepsilon_x, \varepsilon_y)$ are linear coefficients, and

$$a(\varepsilon_x, \varepsilon_y) = \frac{\mu_2(\varepsilon_x, \varepsilon_y) - \mu^2(\varepsilon_x, \varepsilon_y)}{\sigma^2(\varepsilon_x, \varepsilon_y) + \tau}, \quad (8)$$

$$b(\varepsilon_x, \varepsilon_y) = \mu(\varepsilon_x, \varepsilon_y) - a(\varepsilon_x, \varepsilon_y) \mu(\varepsilon_x, \varepsilon_y), \quad (9)$$

where $\mu(\varepsilon_x, \varepsilon_y)$ and $\sigma^2(\varepsilon_x, \varepsilon_y)$ are the mean and variance of L_g , $\mu_2(\varepsilon_x, \varepsilon_y)$ is the mean of L_g^2 in $\omega(x, y)$, τ is the regularization parameter and is set as 0.01.

After using the guided filter, the halo artifacts are significantly reduced. However, the overall appearance of the output image is unsatisfactory due to the low global contrast. To prevent the flat appearance of the guided filter and to improve the performance of our method, we introduce two important factors.

First, the contrast enhancement factor is given by

$$a(x, y) = 1 + \eta \frac{L_g(x, y)}{L_{gmax}}, \quad (10)$$

where η represents the contrast control parameter and is empirically set as 36. L_{gmax} is the maximum output brightness value of the global adaptation. Second, the adaptive nonlinear offset can be written as:

$$\beta = \lambda \bar{L}_g, \quad (11)$$

where λ is a nonlinear control parameter and is empirically set as 10, and $\bar{L}_g$ is the log average luminance of the global adaptation output. By integrating two factors into Eq.(6), the final local adaptive equation is obtained as follows:

$$L_{out}(x, y) = a(x, y) \log\left(\frac{L_g(x, y)}{H_g(x, y)} + \beta\right), \quad (12)$$

where $L_{out}(x, y)$ is the final local adaptive output.

We give a comparison of the effects before and after the introduction of factors, as shown in Fig.3.



Fig.3 Effect comparison with two factors

As mentioned above, using the guided filter causes the flat appearance. The global luminance output $L_g(x, y)$ controls the contrast enhancement factor of each pixel. Since dark pixels make the contrast low, while bright pixels make the contrast high, by introducing contrast enhancement factor, local adaptive output has a more natural appearance than before. And in order to control the nonlinearity of logarithmic function, we propose adaptive nonlinear offset β , which ensures the correct mapping of local adaptive output based on the scene content.

Since the information such as image noise, edges, and details mainly appears in high-pass coefficients, noise can be effectively removed by performing threshold filtering on high-frequency coefficients. Common nonlinear threshold filtering methods include hard threshold filtering, soft threshold filtering and half-threshold filtering^[12]. Although the hard threshold filter retains the details of an image, it will slightly lack smoothness and easily cause visual distortion. The soft threshold filtering method remains relatively smooth in detail and can improve distortion. The half-threshold filtering method has the most effect on detail smoothness, but the algorithm is intricate, and the two thresholds need to be determined, which increases the amount of calculation. Considering these factors, this paper uses a soft threshold filtering method which is expressed as follows:

$$Y = \begin{cases} \text{sign}(X)(|X| - T), & |X| > T \\ X, & |X| \leq T \end{cases}, \quad (13)$$

where Y is the high-frequency coefficient after image denoising, X is the high-frequency coefficient after the image DTCWT, T is the threshold value, $T = \sqrt{2 \log_2(l) \hat{\sigma}}$, where l is the signal length and $\hat{\sigma}$ is the noise variance.

After noise reduction and contrast enhancement in the wavelet domain, we perform an inverse DT-CWT to reconstruct the V-channel, and use a method similar to automatic contrast adjustment to perform white balance correction^[13] on the V-channel. This method can also adjust the contrast of the more obscure image and improve the permeability of the image. After the execution of the algorithm, the image is converted from the HSV space to the RGB color space, and the final enhanced image is obtained.

In order to verify the effectiveness of our proposed algorithm for the enhancement of common low-light images, we collect a large number of images taken at night and low-light images for testing. We use MATLAB 2014a for programming, and use a computer with a four-core 2.50 GHz CPU, 8G RAM, running on Windows 7. The compared algorithms are the traditional MSR algorithm^[7], Fu et al^[14] which is a Retinex-based variational model, Dong et al^[15] which is the algorithm of dark primary color theory and Wang et al^[16] which is an adaptive contrast improvement and contrast-related saturation adjustment algorithm. The test images are shown as Fig.4.

Subjective results are shown in Fig.5. Moreover, three evaluation criteria are adopted. They are discrete *Entropy*^[17], *Clarity* and *Contrast*. Tabs.1—3 show the results.

Entropy, the amount of information in an image, is calculated as:

$$H(p) = \sum_{i=0}^{L-1} p(i) \log_2 p(i), \quad (14)$$

where p is the probability density function. The larger the entropy value, the more details contained in the image, and the better the visual effect of the image.

Clarity is an important indicator of the pros and cons of image quality, and refers to the sharpness of the image seen by human eyes. The definition of the sharpness as the spatial frequency of the image reflects the overall activity of the image. The larger the value, the better the image effect. Supposing the image size is $M \times N$, the definition formula is:

$$Q = \sqrt{RF^2 + CF^2}, \quad (15)$$

where RF is the line frequency of the image and CF is the column frequency of the image, which is calculated as:

$$RF = \sqrt{\frac{1}{(M-1)N} \sum_{i=1}^M \sum_{j=1}^N [I_{i,j+1} - I_{i,j}]^2}, \quad (16)$$

$$CF = \sqrt{\frac{1}{(M-1)N} \sum_{i=1}^M \sum_{j=1}^N [I_{i+1,j} - I_{i,j}]^2}. \quad (17)$$

Contrast refers to the measurement of the different brightness levels between the brightest white and the darkest black in the light and dark areas of an image. A larger difference indicates a greater contrast, and a smaller difference indicates a smaller contrast. The greater the contrast, the clearer the image and the brighter the colors. The formula is:

$$C = \sum_{\delta} \delta(i, j)^2 P_{\delta}(i, j), \quad (18)$$

where $\delta(i, j) = |i - j|$ is the grayscale difference between adjacent pixels of the image, and $P_{\delta}(i, j)$ is the pixel distribution probability that the grayscale difference between adjacent pixels of the image is δ . In this paper, we use the ratio of contrast between the enhanced image and the original image, that is, the degree of change in the contrast of the image before and after enhancement.



Fig.4 Test images

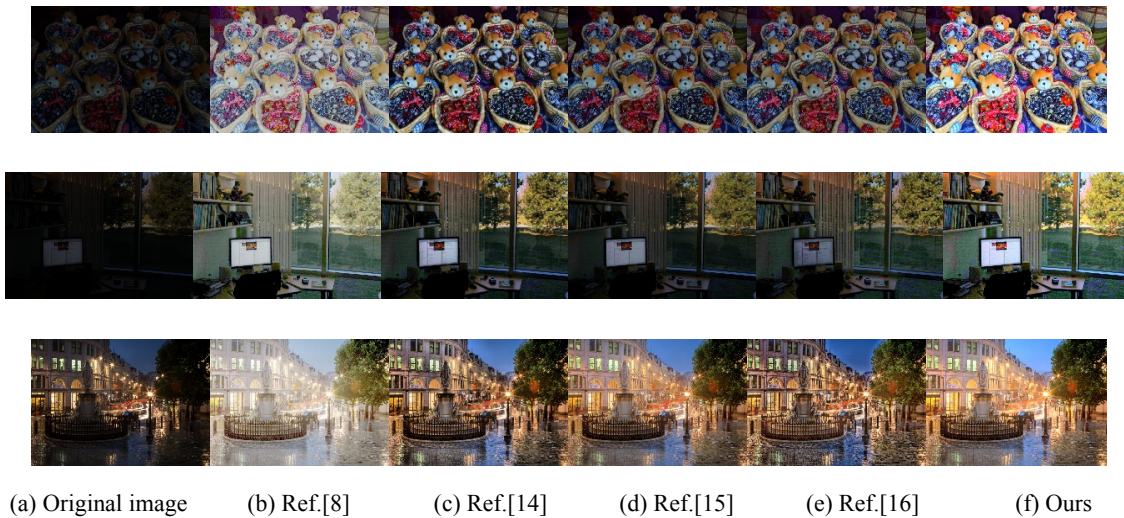


Fig.5 Comparison of various methods

Tab.1 Comparison of Entropy

Image/Method	Ref.[7]	Ref.[14]	Ref.[15]	Ref.[16]	Ours
Barrel	7.784	7.349	7.408	7.489	7.643
Horse	7.205	7.014	6.960	7.260	6.906
Toys	7.547	7.584	7.222	7.332	7.914
Hallway	7.061	6.921	6.753	6.747	7.584
Desk	7.335	6.994	6.590	6.825	7.521
Frame	7.381	7.122	6.375	6.877	7.772
Street	7.551	7.717	7.750	7.695	7.770
Bridge	7.472	7.429	7.522	7.284	7.711

Tab.2 Comparison of Clarity

Image/Method	Ref.[7]	Ref.[14]	Ref.[15]	Ref.[16]	Ours
Barrel	8.320	8.435	7.193	8.833	8.615
Horse	5.398	5.015	4.060	5.433	5.108
Toys	7.282	7.290	4.922	5.776	9.122
Hallway	5.435	6.252	5.370	4.154	5.712
Desk	9.237	8.747	6.260	7.137	10.136
Frame	9.479	9.514	5.957	8.063	13.787
Street	8.322	13.453	10.968	15.599	10.269
Bridge	4.523	4.708	4.423	6.033	4.676

Tab.3 Comparison of Contrast

Image/Method	Ref.[7]	Ref.[14]	Ref.[15]	Ref.[16]	Ours
Barrel	0.964	1.249	0.999	1.151	1.302
Horse	0.661	0.603	0.471	0.355	0.664
Toys	3.149	3.053	1.590	1.992	4.488
Hallway	0.358	0.345	0.233	0.307	0.671
Desk	0.893	0.806	0.388	0.689	1.358
Frame	1.613	1.503	0.612	1.015	2.976
Street	0.956	2.239	1.689	2.879	2.381
Bridge	0.398	0.292	0.196	0.411	0.332

From the comparison of experimental results of various algorithms, it can be seen that each algorithm enhances the image to some extent. However, in the enhancement process of MSR, an image may distort the detail of the image due to the increased noise, the real color of the object cannot be displayed, and the overall visual effect becomes worse. Fu et al^[14] proposed the algorithm based on the variational model of Retinex theory, then there will be some image processing step effect, resulting in the image of the local color uncoordinated. After an image is processed by Dong et al^[15], the image will be darkened and the color distortion will appear easily. The Wang et al^[16] algorithm can effectively improve the visibility of the image, but some

images have halo artifacts on the edges with high contrast. From the results of algorithm processing and the test data of each quality evaluation index, our proposed algorithm effectively improves the contrast of an image to a certain extent, makes the image look more natural and at the same time effectively improves the readability of the low-light image.

Images captured in dark conditions have a narrower dynamic range, lower contrast, and high-level noise. Considering the features of low-light images, we propose a low-light image enhancement algorithm based on Retinex theory and dual-tree complex wavelet transform. Since the wavelet theory has superior performance in noise reduction, we use DT-CWT to process the V-channel of the image. According to the features of the wavelet coefficients in the wavelet domain, we then perform contrast enhancement and noise reduction respectively, and reconstruct the V-channel using the white balance algorithm. Finally, we convert the enhanced image back to the RGB color space to get the final enhanced image. Experimental results verify that the proposed method can effectively improve the contrast and brightness of the image. It retains a lot of details and can obtain more natural visual effects than other algorithms.

References

- [1] Pooja Bidwai and D. J. Tuptewar, IEEE International Conference on Information Processing, 511 (2015).
- [2] Tingting Sun and Cheolkon Jung, IEEE International Conference on Acoustics, Speech and Signal Processing, 1741 (2016).
- [3] Seonhee Park, Soohwan Yu, Byeongho Moon, Seungyong Ko and Joonki Paik, IEEE Transactions on Consumer Electronics **63**, 178 (2017).
- [4] Rajasekhar Karumuri and Rajasekhar Karumuri, IEEE International Conference on Communication and Electronics Systems, 545 (2017).
- [5] Xiaojie Guo, Yu Li and Haibin Ling, IEEE Transactions on Image Processing **26**, 982 (2017).
- [6] Kim Kyungil, Soohyun Kim and Kyung-Soo Kim, IET Image Processing **12**, 465 (2018).
- [7] Jobson Daniel J., Zia-urRahman and Glenn A. Woodell, IEEE Transactions on Image Processing **6**, 451 (1997).
- [8] Jobson Daniel J., Zia-urRahman and Glenn A. Woodell, IEEE Transactions on Image processing **6**, 965 (1997).
- [9] He Kaiming, Jian Sun and Xiaoou Tang, IEEE Transactions on Pattern Analysis and Machine Intelligence **35**, 1397 (2013).
- [10] Selesnick I W, Baraniuk R G and Kingsbury N C, IEEE Signal Processing Magazine **22**, 123 (2005).
- [11] Drago F, Myszkowski K, Annen T and Chiba N, Computer Graphics Forum **22**, 419 (2003).
- [12] Mo Wei Jian, Bai Hui Zhu and Zhi Ping Wan, IEEE International Conference on Computational and Information Sciences, 171 (2013).
- [13] Nicolas Limare, Jose-Luis Lisani, Jean-Michel Morel,

- Ana Belén Petro and Catalina Sbert, Image Processing On Line **1**, 297 (2011).
- [14] Xueyang Fu, Ye Sun, Minghui LiWang, Yue Huang, Xiaoping Zhang and Xinghao Ding, IEEE International Conference on Acoustics, Speech and Signal Processing, 1190 (2014).
- [15] Xuan Dong, Guan Wang, Yi Pang, Weixin Li, Jiangtao Wen, Wei Meng and Yao Lu, IEEE International Conference on Multimedia and Expo, 1 (2011).
- [16] Shuhang Wang, Woon Cho, Jinbeum Jang, Mongi A. Abidi and Joonki Paik, Journal of the Optical Society of America A **34**, 7 (2017).
- [17] Shannon C E, Bell System Technical Journal, 379 (1948).